

PA Instruction

DSC 291/190: Trustworthy Machine Learning, SP'26

Instructor: Dr. Lily Weng

Instructions

- You should do this PA individually. Collaboration is not allowed.
- The homework has to be submitted on Gradescope before the submission deadline.
- The submission deadline is **Thursday, April 23rd, 2026 at 11:59 PM**. Delay per day gives a **20% penalty**. If we do not receive your submission by **Tuesday, April 28th at 11:59 AM**, it will be graded as 0 pt.

HW Structure:

- This homework consists of 2 questions. The first question deals with **Vision Model Training and Evaluation** and the second questions deals with **Language Model Training and Evaluation**. Each part has its own dedicated Colab notebook which you can find in [this zip](#).
- The two parts and their subparts are as follows:
 - a. Vision Model Training and Evaluation
 - Basic Vision Models [Experiments with MNIST] (35 points)
 - Advanced Vision Models [Experiments with CIFAR10] (25 points)
 - Training a Robust Model (10 points) – **Optional/Bonus**
 - b. Language Model Training and Evaluation
 - Train Tiny-GPT model from scratch on Salesforce/wikitext dataset (40 points)
- Further instructions for each of the questions are given in their respective notebooks.

Submission Format:

- For each question:
 - You may choose to either use Colab or download the file as .ipynb and run it on your local machine as a Jupyter notebook.
 - Once you have completed your solutions, save the notebook as a PDF.
 - If using Colab do: Download the notebook from colab as a .ipynb file (File -> Download as -> .ipynb) and use `jupyter nbconvert --to PDF filename.ipynb --output output\_filename.pdf` in the terminal.
 - Alternatively, you can open the .ipynb file in Jupyter and follow the next bullet point.
 - If using Jupyter do: *File -> Download as -> PDF via LaTeX*
 - **Make sure all your solutions are displayed in the pdf.** It's okay if the provided questions get cut off.

- Look at the PDF file and make sure all your solutions are there, displayed correctly. The PDF is the only thing the grader can see.
- Once the PDF files are ready, merge them into one large PDF.
- Submit the final PDF on Gradescope individually.
 - In Gradescope, assign the pages of the solution pdf to the corresponding question number, as shown in [this video tutorial](#) or as shown in the screenshot below.

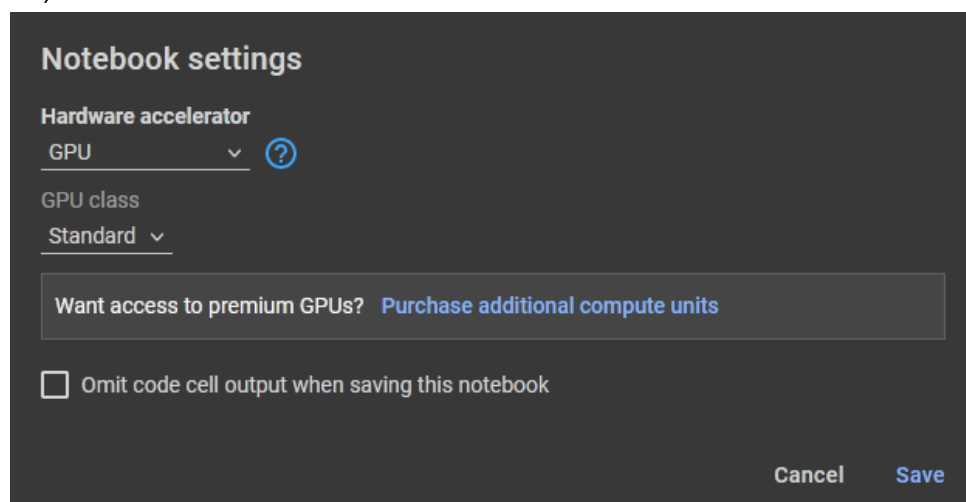
Question Outline
Select pages to assign to Questions 1, 2.1, and 2.2.

TITLE	POINTS
1 Chapter 1 ex 3 P1 x	4.0 pts
2 Chapter 1 ex 7	8.0 pts
2.1 7a P1 x	4.0 pts
2.2 7b P1 x	4.0 pts
3 Chapter 2 ex 5	4.0 pts

- Note that PDF of the solved notebooks is the only submission format. Do not submit python (.py) or notebook (.ipynb) extension files.
 - **Note that we do want you to show your code in the pdf (please don't collapse the cells in Colab so that we can see and grade them).**

Enable GPU in Colab

- We highly recommend using a GPU in Colab for faster runtimes. To enable a GPU, go to Runtime → Change Runtime Type and select “GPU” in the dropdown list (as shown below) and click “Save”.



- Note that there is a usage limit per day on Colab, so please test your code first without GPU (selecting “None” in the above dropdown list) and run your final code on GPU.
- Also note that Colab will disconnect the session if there is inactivity in the browser tab.

Colab Pro for Students

Pay As You Go	Recommended Colab Pro	Colab Pro+	Colab Enterprise
\$9.99 for 100 Compute Units \$49.99 for 500 Compute Units	\$9.99 per month Colab Pro for Education No cost for students and educators	\$49.99 per month	Pay for what you use
✓ No subscription required. Only pay for what you use. ✓ Faster GPUs Upgrade to more powerful GPUs.	✓ 100 compute units per month Compute units expire after 90 days. Purchase more as you need them. ✓ Faster GPUs Upgrade to more powerful GPUs. ✓ More memory Access our highest memory machines.	All of the benefits of Pro, plus: ✓ An additional 500 compute units per month, totaling 600 compute units per month. Compute units expire after 90 days. Purchase more as you need them. ✓ Faster GPUs Priority access to upgrade to more powerful premium GPUs. ✓ Background execution With compute units, your actively running notebook will continue running for up to 24hrs, even if you close your browser.	✓ Integrated Tightly integrated with Google Cloud services like BigQuery and Vertex AI. ✓ Enterprise notebook storage Replace your usage of Google Drive notebooks with GCP notebooks, stored and shared within your cloud console. ✓ Productive Generative AI powered code completion and generation.

- Students may use **Colab Pro / Colab Pro for Education** for improved compute availability, access to faster GPUs, and high-memory runtimes. For eligible students, Colab Pro for Education is at no cost and includes the same benefits as a standard Colab Pro subscription for one year.